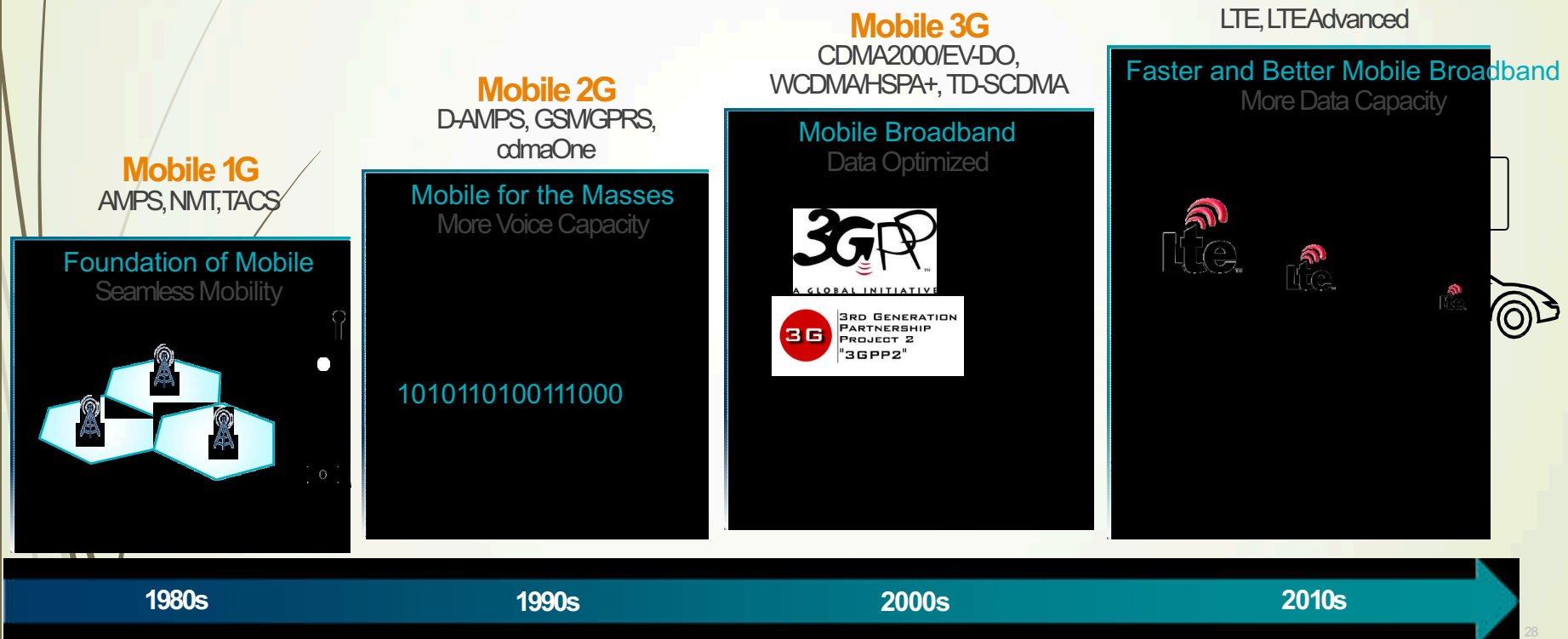


4G LTE is the First All-IP Cellular Network



Agenda

- 5G Cellular Networks
- Evolution of Cellular Networks
- Network Architecture
- Network Operations and Protocol Stacks

Standardization Body: 3GPP (3rd Generation Partnership Project)

- Evolves and standardizes GSM, UMTS, LTE among others

“The 3rd Generation Partnership Project (3GPP) unites [Six] telecommunications standard development organizations (ARIB, ATIS, CCSA, ETSI, TTA, TTC), known as “Organizational Partners” and provides their members with a stable environment to produce the highly successful Reports and Specifications that define 3GPP technologies”

Some Other Standardization Bodies

Generation	3GPP Circuit Switched	3GPP Packet Switched	3GPP2	Wimax Forum
2G	GSM		cdmaOne	
2.5G		GPRS		
2.75G		EDGE		
3G	UMTS		CDMA2000	
3.5G		HSPA/+	CDMA EV-DO	
4G		LTE	UMB	WiMAX

LTE (Long Term Evolution) Networks

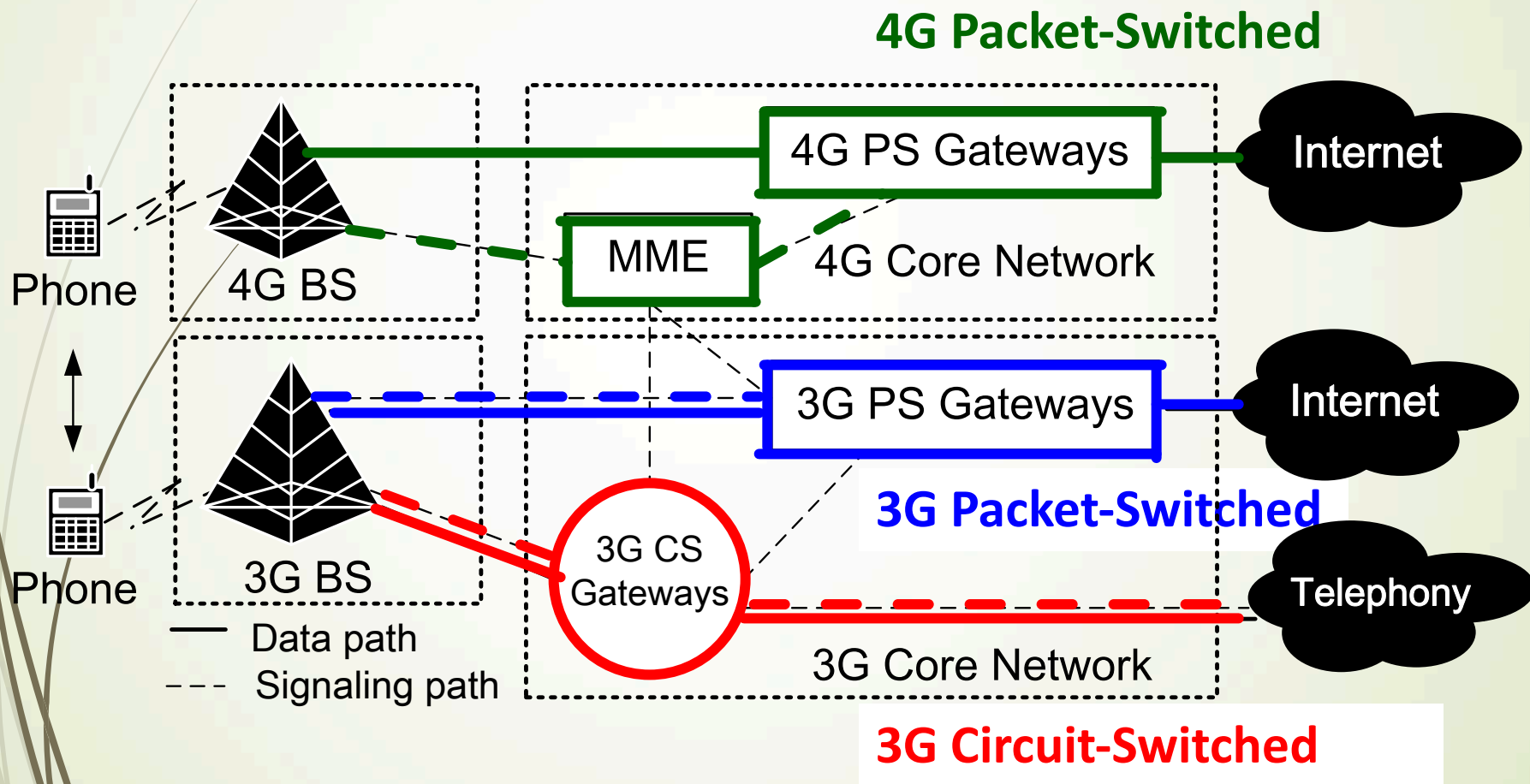
- The 4G standard that is deployed worldwide
- 4G LTE:
 - Increased speed
 - All-IP network (no more circuits)
 - New air interface: OFDMA (Orthogonal Frequency-Division Multiple Access), MIMO (multiple antennas)
 - New service paradigm (e.g., VoLTE: voice server over IP networks)
- LTE is always evolving and 3GPP often has new “releases”
 - First release: Rel-8
 - Current: Rel-11, Rel-12
 - Toward LTE-Advanced (4.5G)

Q: VoLTE is similar to VoIP. Why do we need it?

2G Network Architecture (GSM)



3G/4G Network Architecture



MME: Mobility Management Entity
BS: Base Station (4G: eNodeB)

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